

RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)



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Each of the following case study questions can be answered using a single SQL statement:

1. What is the total amount each customer spent at the restaurant?
2. How many days has each customer visited the restaurant?
3. What was the first item from the menu purchased by each customer?
4. What is the most purchased item on the menu and how many times was it purchased by all customers?
5. Which item was the most popular for each customer?



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

6. Which item was purchased first by the customer after they became a member?
7. Which item was purchased just before the customer became a member?
8. What is the total items and amount spent for each member before they became a member?
9. If each 1 Taka spent equates to 10 points and kacchi has a 2x points multiplier - how many points would each customer have?
10. In the 3 days after a customer joins the program((including their join date)
 1. they earn 2x points on all items, not just Kacchi(Only for first 3 days)
 2. After 3 days - If each 1 Taka spent equates to 10 points and kacchi has a 2x points multiplierhow many points do customer A and B have at the end of January, 2021?



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1  --1. What is the total amount each customer spent at the restaurant?
2  --sales,menu
3  ✓ select customer_id, sum(price) as "Total Spent"
4     from sales as s
5     inner join menu as me
6     on me.product_id = s.product_id
7     group by customer_id;
```

Data Output Messages Notifications

	customer_id character varying (10) 🔒	Total Spent bigint 🔒
1	B	1960
2	C	600
3	A	1660



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1  --2. How many days has each customer visited the restaurant?
2  --salse
3  ✓ select customer_id, count(order_date) as "Visited Days"
4     from sales
5     group by customer_id;
```

Data Output Messages Notifications

	customer_id character varying (10) 🔒	Visited Days bigint 🔒
1	B	10
2	C	3
3	A	9



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1  -- 3. What was the first item from the menu purchased by each customer?
2  -- sales,menu
3  -- order_date,product_name,customer_id
4  v select distinct customer_id,product_name
5      from
6
7  (select order_date,product_name,customer_id,
8      DENSE_RANK() over
9      (partition by customer_id order by order_date) as rnk
10     from sales as s
11     inner join menu as me
12     on s.product_id = me.product_id) as tt
13 where rnk = 1;
```

Data Output Messages Notifications

	customer_id character varying (10) 🔒	product_name character varying (50) 🔒
1	A	kacchi
2	A	roast
3	B	roast
4	C	beef



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1  -- 4. What is the most purchased item on the menu and how many times was it purchased by all customers
2  v select customer_id, count(*)
3      from
4      (select s.product_id, me.product_name, count(*)
5       from sales as s
6         inner join menu as me
7          on s.product_id = me.product_id
8      group by s.product_id, me.product_name
9      order by count(*) desc
10     limit 1
11  ) as tt
12  inner join sales as s on s.product_id = tt.product_id
13  group by customer_id;
```

Data Output Messages Notifications

	customer_id character varying (10)	count bigint
1	A	4
2	B	3
3	C	3



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1 --5. Which item was the most popular for each customer?
2 -- sales ,menu
3 v select * from
4 (
5   select *,
6     dense_rank() over(partition by customer_id order By ProductCount desc) as rnk
7   from
8   (
9     select s.customer_id,me.product_name,count(*) as ProductCount
10    from
11      sales as s inner join menu as me on me.product_id = s.product_id
12    group by s.customer_id,me.product_name
13   ) as t1
14   ) as t2
15  where rnk = 1;
```

Data Output Messages Notifications



customer_id	product_name	productcount	rnk
character varying (10)	character varying (50)	bigint	bigint
A	beef	4	1
B	kacchi	4	1
C	beef	3	1

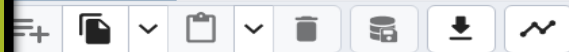
Total rows: 3 of 3 Query complete 00:00:00.092



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1  -- 6. Which item was purchased first by the customer after they became a member?|
2  v select customer_id,product_name
3      from
4      (
5      select *,
6          dense_rank() over(partition by tt.customer_id order by tt.order_date asc ) as rnk
7      from
8      (
9      select s.customer_id,s.order_date,me.product_name
10         from sales as s
11        inner join members as mb on s.customer_id = mb.customer_id
12        inner join menu as me on s.product_id = me.product_id
13       where order_date>join_date)as tt
14         ) as t3
15     where rnk = 1;
```

Data Output Messages Notifications



	customer_id character varying (10) 🔒	product_name character varying (50) 🔒
1	A	roast
2	B	kacchi



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1  -- 7. Which item was purchased just before the customer became a member?
2  v select customer_id,product_name
3      from
4      (
5      select *,
6          dense_rank() over(partition by tt.customer_id order by tt.order_date desc ) as rnk
7      from
8      (
9      select s.customer_id,s.order_date,me.product_name
10         from sales as s
11         inner join members as mb on s.customer_id = mb.customer_id
12         inner join menu as me on s.product_id = me.product_id
13         where order_date<join_date)as tt
14         ) as t3
15     where rnk = 1;
```

Data Output Messages Notifications



	customer_id character varying (10)	product_name character varying (50)
1	A	kacchi
2	A	roast
3	B	kacchi

Total rows: 3 of 3 Query complete 00:00:00.063

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RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
1 -- 8. What is the total items and amount spent for each member before they became a member?  
2  
3  
4 v select s.customer_id, count(s.order_date) as "Total item" ,sum(me.price) as "Total Amount"  
5     from sales as s  
6     inner join members as mb on s.customer_id = mb.customer_id  
7     inner join menu as me on s.product_id = me.product_id  
8     where order_date < join_date  
9     group by s.customer_id  
10  
11
```

Data Output Messages Notifications



customer_id character varying (10) 🔒	Total item bigint 🔒	Total Amount bigint 🔒
B	4	610
A	2	370



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

-- 9. If each 1 Taka spent equates to 10 points and kacchi has a 2x points multiplier - how many point

```
select s.customer_id,  
       sum(case when me.product_id = 1 then price*20 else price*10 end) as points  
from sales as s  
inner join menu as me  
on me.product_id = s.product_id  
group by customer_id
```

Data Output Messages Notifications

customer_id	points
character varying (10)	bigint
B	29600
C	6000
A	21600



RESTAURANT DATA ANALYSIS (KACCHI_BD.COM)

```
7 v select customer_id,  
8       sum(case  
9         when rnk between 1 and 3 then price*20  
10        when rnk not between 1 and 3 and product_id = 1 then price*20  
11        else price*10 end) as points from  
12 (select  
13   s.*,me.price,  
14   dense_rank() over(partition by s.customer_id order by order_date ) as rnk  
15   from sales as s  
16   inner join members as mm on s.customer_id = mm.customer_id  
17   inner join menu as me on s.product_id = me.product_id  
18   where s.order_date >= mm.join_date  
19   and s.order_date >= '2021-01-01' and s.order_date < '2021-02-01')as tt  
20 group by customer_id  
21
```

Data Output Messages Notifications

	customer_id character varying (10)	points bigint
1	A	15800
2	B	23000

